

DBMS PHASE 3

NORMALIZATION

Group-13

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UNIVERSAL TABLE -COLLEGE

Attribute	Row 1	Row 2	Row 3
student_id	S001	S001	S002
stud_name	John Doe	John Doe	Jane Smith
dob	2000-05-15	2000-05-15	1999-08-22
email	john.doe@univ.edu	john.doe@univ.edu	jane.smith@univ.edu
department	CS	CS	EE
advisor_id	F001	F001	F002
student_phone	1234567890	9876543210	1122334455
faculty_id	F001	F001	F002
fac_name	Alice Johnson	Alice Johnson	Bob Brown
fac_email	alice.j@univ.edu	alice.j@univ.edu	bob.brown@univ.edu
designation	Professor	Professor	Assistant Prof
faculty_department	CS	CS	EE
faculty_phone	5551234567	5551234567	5559876543
application_id	A001	A001	A002
leave_type	Sick Leave	Sick Leave	Vacation

start_date	2024-01-10	2024-01-10	2024-02-01
end_date	2024-01-15	2024-01-15	2024-02-07
purpose	Recovery from illness	Recovery from illness	Family visit
status	Approved	Approved	Pending
submission_date	2024-01-05	2024-01-05	2024-01-25
extended_request	No	No	Yes
proof_id	P001	P001	P002
proof_type	Medical Report	Medical Report	Travel Itinerary
file_path	/proofs/med_report.pdf	/proofs/med_report.pdf	/proofs/travel_doc.pdf
upload_date	2024-01-05	2024-01-05	2024-01-24

ATTRIBUTES OF THE UNIVERSAL TABLE :

student_id	Unique identifier for the student.
stud_name	Student's name.
dob	Date of birth of the student.
email	Student's email address.
department	Department to which the student belongs.
advisor_id	Faculty ID of the student's advisor.
student_phone	Phone number(s) of the student.
faculty_id	Unique identifier for the faculty.
fac_name	Faculty's name.
fac_email	Faculty's email address.
designation	Faculty's designation (e.g., Professor, Assistant Prof).
faculty_department	Department to which the faculty belongs.
faculty_phone	Phone number(s) of the faculty.
application_id	Unique identifier for the leave application.

leave_type	Type of leave (e.g., Sick Leave, Vacation).
start_date	Start date of the leave.
end_date	End date of the leave.
purpose	Purpose of the leave application.
status	Status of the leave application (e.g., Approved, Pending).
submission_date	Date when the leave application was submitted.
extended_request	Whether an extended leave was requested (Yes/No).
proof_id	Unique identifier for the supporting proof document.
proof_type	Type of proof document (e.g., Medical Report, Travel Itinerary).
file_path	Path to the uploaded proof document.
upload_date	Date when the proof document was uploaded.

CONVERSION TO 1NF

COMPOSITE ATTRIBUTE

STUD_NAME, FAC_NAME

MULTI VALUED ATTRIBUTE

STUD-PHONE

NORMALISING

- Creating a new table for composite and multivalued attributes

COLLEGE (student_id, stud_first_name, stud_last_name, dob, email, department,

advisor_id, faculty_id, fac_first_name, faculty_last_name, fac_email, designation,

faculty_department, application_id, leave_type, start_date, end_date, purpose,

status, submission_date, extended_request, proof_id, proof_type, file_path,
upload_date)

-----splitted stud_name to stud_first_name and stud_last_name

Stud_ph(student_id, student_phone)

FUNCTIONAL DEPENDENCY

PRIMARY KEYS ARE :

student_id, faculty_id, application_id, proof_id for COLLEGE table

student_id for STUD_PH table

COLLEGE TABLE(FD)

stud_id -> { student_id, stud_first_name, stud_last_name, dob, email, department}
stud_id, faculty_id -> { advisor_id }

faculty_id -> { fac_first_name, fac_last_name, fac_email, designation,
faculty_department }

application_id -> { leave_type, start_date, end_date,
status, extended_request, purpose, submission_date }

proof_id -> { proof_type, file_path, upload_date }

FOR STUD PH TABLE(FD)

student_id -> stud_phone

CONVERSION TO 2NF

- It is in 1 nf
- Have partial dependency , so converting such there is no partial dependency

CONVERTED TO RELATIONS WITH PARTIAL PRIMARY KEY AS THE PRIME TTRIBUTES

Student(student_id,stud_first_name,stud_last_name,dob, email, department)

Advisor(stud_id,faculty_id,advisor_id)

Faculty(faculty_id,fac_first_name,fac_last_name, fac_email, designation,faculty_department)

Application(application_id,leave_type, start_date, end_date, status,extended_request ,purpose,submission_date)

Proof(proof_id, proof_type, file_path,upload_date)

Stud_ph(stud_id,stud_phone)

FUNCTIONAL DEPENDENCIES

- FOR STUDENT TABLE

stud_id -> { student_id,stud_first_name,stud_last_name,dob, email, department}
stud_id,faculty_id-> {advisor_id}

- FACULTY TABLE

faculty_id->{ fac_first_name,fac_last_name, fac_email, designation, faculty_department }

- APPLICATION TABLE

application_id->{ leave_type, start_date, end_date, status,extended_request ,purpose,submission_date}

- PROOF TABLE

proof_id->{ proof_type, file_path,upload_date }

- FOR STUD_PH TABLE

student_id->stud_phone

RIGHT NOW IT IS IN 2NF

COVERTING TO 3NF

- It is in 2NF
- Not have transitive dependency

So it is in 3NF

CHECKING IT IS IN BCNF

It is in BCNF since every determinant is a primary a.k.a candidate key

SO FINAL TABLES :

Student(student_id,stud_first_name,stud_last_name,dob, email, department)

Advisor(stud_id,faculty_id,advisor_id)

Faculty(faculty_id,fac_first_name,fac_last_name, fac_email, designation,faculty_department)

Application(application_id,leave_type, start_date, end_date, status,extended_request ,purpose,submission_date)

Proof(proof_id, proof_type, file_path,upload_date)

Stud_ph(stud_id,stud_phone)